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CIS344
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Final Project Documentation

The purpose of this project was to design and build a Real Estate Agency Portal using PHP and MySQL. The goal of the system is to allow agents, buyers, and renters to interact with property listings through a simple web-based platform. Agents can add and manage property listings, while buyers and renters can browse available properties, save favorites, and send inquiries about homes they are interested in. This project was created using PHP, MySQL, phpMyAdmin, HTML, and CSS.

The database for the project was built in MySQL and connected to the PHP pages through a database connection file. Several tables were created to keep the system organized, including Users, Properties, Inquiries, Transactions, and Favorites. The Users table stores account information such as usernames, passwords, contact information, and user roles. The Properties table stores information about each property including the address, city, price, and availability status. The Inquiries table allows buyers and renters to send messages about properties, while the Transactions table keeps track of completed sales and rentals. The Favorites table gives users the ability to save properties they may want to come back to later.

Relationships between the tables were created using primary keys and foreign keys so all the data stays connected properly. For example, each property is connected to an agent through the agentId field, while inquiries and transactions connect users with specific properties. Sample data was added into the tables to test the relationships and make sure everything was working correctly. On the PHP side, the platform includes features such as secure login, adding properties, viewing listings, submitting inquiries, and retrieving user information. Passwords are protected using password_hash() and checked using password_verify() for better security. Session handling and role-based access control were also implemented so agents, buyers, and renters each have access to different parts of the system.

The platform was tested by creating users, adding properties, submitting inquiries, saving favorites, and processing transactions. Different login roles were also tested to make sure permissions and features worked correctly. During development, there were some challenges such as database connection problems and stored procedure compatibility issues with MariaDB, but these issues were fixed by adjusting the SQL syntax and configuration settings. Overall, this project was a good experience for learning how PHP and MySQL work together in a real-world application. It helped demonstrate important concepts such as database relationships, and secure login systems.